

S. V. National Institute of Technology (SVNIT), Surat-07, Gujarat, India
Department of Electronics Engineering

V22E1004

Mid Semester Examination (September-2024)

B. Tech - III, 5th Sem

Total Duration : 90 Minutes

Total Marks : 30

EC301 - Digital Signal Processing

Max Allowed Marks : 30

Instruction: (01) Attempt all questions. & Maintain proper sequence.

(02) Assume necessary Data if required.

(03) Figure to the right indicates contains mark.

1. (a) The z-transform $H(z)$ of the system is given by;

$$H(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{1}{1 - 2z^{-1}} + \frac{1}{1 - 5z^{-1}}$$

Comment on the causality and stability of the system.

- (b) Determine the unit step response for the system described by the second-order difference equation, while considering, it is in initially at relaxed condition;

$$y(n) - 4y(n-1) + 4y(n-2) = x(n) - x(n-1)$$

2. (a) Consider 28-point DFT. How many computations required for that, if direct-point DFT algorithm is used. What a percentage improvement in computation of same point DFT, if the radix-2 FFT algorithm is used. Also find the percentage improvement if same point DFT is done using split-radix algorithm.

- (b) The DFT of a 5-point signal $x(n)$, $0 \leq n \leq 4$ is $X(k) = \{5, 6, 1, 2, 9\}$ for $0 \leq k \leq 4$.

A new signal $g(n)$ is defined by, $g(n) = W_5^{-2n}x(n)$ for $0 \leq n \leq 4$.What are the DFT coefficients $G(k)$ of the signal $g(n)$, for $0 \leq k \leq 4$?

- (c) Explain Decimation-in-time radix-2 FFT algorithm for calculation of Inverse Discrete Fourier Transform. Also draw the butterfly diagram with detail explanation of each step. How many multiplier and Adder is needed to perform above operation.

3. (a) Determine the particular solution of the difference equation, when the forcing function, $x(n) = 2^n u(n)$;

$$y(n) = \frac{5}{6}y(n-1) - \frac{1}{6}y(n-2) + x(n)$$

- (b) The frequency response of LTI system is given by,

$$H(e^{j\omega}) = \frac{1 - \frac{1}{4}e^{-j\omega}}{1 - \frac{1}{2}e^{-j\omega/2} - \frac{1}{2}e^{-j\omega}}$$

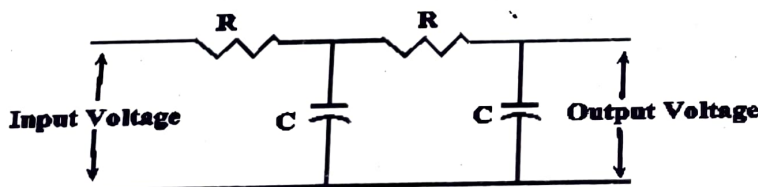
Find out the frequency response of its inverse system. What is behaviour of inverse system, if unit step is applied at the input of this system.

4. (a) The system function is realized using following input-output relation function as,

$$2y(n) + y(n-2) - 3y(n-3) = x(n) - x(n-1) + 2x(n-2) - 3x(n-4)$$

Realize its structure using the delay block, adder and multiplier. What is overall DC gain of system. Find out frequency response of a system. Comment on phase response this system (i.e linear or non-linear)?

- (b) For the given RC filter circuit, (i) Calculate the transfer function. (ii) Identify whether the circuit is low-pass or high-pass or band-pass or any other filter. (iii) What is the cut-off frequency of the circuit?



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